Decimal Operations

Line up the points. Then think in tenths and hundredths.

Add and subtract: line up the points

$$\begin{array}{r} 12.50 \\ +\ \ 3.75 \\ \hline 16.25 \end{array}$$

12.5 is 12.50: write the zero so every column is filled.
Then add like whole numbers. The point drops straight down.

Multiply: whole numbers first, then name the piece

2.4×1.3: $24 \times 13 = 312$ $\qquad$ $2.4 \times 1.3 = 3.12$

Tenths times tenths are hundredths, so 312 hundredths = 3.12.

Divide: make the divisor a whole number

$4.68 \div 0.6 = 46.8 \div 6 = 7.8$

Both numbers × 10: the answer stays the same. Then divide by a whole number.

Your turn!

$3.6 \times 0.4 =$ 1.44

For teachers

Goal: add, subtract, multiply and divide multi-digit decimals (6.NS.3). Say “Line up the points” and “What size are the pieces?” Never just “count the decimal places”: name tenths and hundredths.

Decimal Operations: Practice

Line up the points. Name the pieces.

Part 1 Line up the points Write the gray zeros if they help.

1)
$$\begin{array}{r} 23.60 \\ +\ 8.75 \\ \hline 32.35 \end{array}$$

2)
$$\begin{array}{r} 40.20 \\ -\ 17.48 \\ \hline 22.72 \end{array}$$

3)
$$\begin{array}{r} 6.08 \\ +\ 19.70 \\ \hline 25.78 \end{array}$$

4)
$$\begin{array}{r} 52.30 \\ -\ 6.84 \\ \hline 45.46 \end{array}$$

5)
$$\begin{array}{r} 315.40 \\ +\ 8.29 \\ \hline 323.69 \end{array}$$

6)
$$\begin{array}{r} 70.05 \\ -\ 36.80 \\ \hline 33.25 \end{array}$$

7)
$$\begin{array}{r} 9.40 \\ +\ 126.37 \\ \hline 135.77 \end{array}$$

8)
$$\begin{array}{r} 83.10 \\ -\ 45.66 \\ \hline 37.44 \end{array}$$

9)
$$\begin{array}{r} 204.70 \\ +\ 3.58 \\ \hline 208.28 \end{array}$$

Part 2 Add and subtract No help this time.

10) $38.27 + 4.6 =$ **42.87**

11) $61.4 - 27.85 =$ **33.55**

Decimal Operations: Practice (continued)

Part 2 Add and subtract (continued)

12) $7.09 + 16.8 =$ **23.89**

13) $100.3 - 45.67 =$ **54.63**

14) $245.6 + 18.75 =$ **264.35**

15) $32.06 - 9.4 =$ **22.66**

16) $0.86 + 13.7 =$ **14.56**

17) $508.2 - 136.45 =$ **371.75**

18) $71.4 + 29.68 =$ **101.08**

19) $44.1 - 6.37 =$ **37.73**

20) $153.07 + 8.9 =$ **161.97**

21) $90.6 - 52.78 =$ **37.82**

22) $26.35 + 119.8 =$ **146.15**

23) $17.2 - 8.46 =$ **8.74**

Decimal Operations: Practice (continued)

Part 2 Add and subtract (continued)

24) $4.75 + 63.9 =$ **68.65**

Part 3 Multiply: whole numbers first Then name the pieces.

25) 2.7×1.4: $27 \times 14 =$ **378** so $2.7 \times 1.4 =$ **3.78**

26) 3.6×2.3: $36 \times 23 =$ **828** so $3.6 \times 2.3 =$ **8.28**

27) 0.8×4.7: $8 \times 47 =$ **376** so $0.8 \times 4.7 =$ **3.76**

28) 5.2×1.9: $52 \times 19 =$ **988** so $5.2 \times 1.9 =$ **9.88**

29) 1.25×3.4: $125 \times 34 =$ **4250** so $1.25 \times 3.4 =$ **4.25**

30) 6.4×0.35: $64 \times 35 =$ **2240** so $6.4 \times 0.35 =$ **2.24**

31) 4.8×2.6: $48 \times 26 =$ **1248** so $4.8 \times 2.6 =$ **12.48**

32) 0.72×1.5: $72 \times 15 =$ **1080** so $0.72 \times 1.5 =$ **1.08**

33) 7.3×4.1: $73 \times 41 =$ **2993** so $7.3 \times 4.1 =$ **29.93**

34) 2.15×6.2: $215 \times 62 =$ **13330** so $2.15 \times 6.2 =$ **13.33**

Decimal Operations: Practice (continued)

Part 4 Multiply No help this time.

35) $3.25 \times 4 = 13$

36) $6.7 \times 8.2 = 54.94$

37) $0.45 \times 2.8 = 1.26$

38) $12.6 \times 3.5 = 44.1$

39) $9.4 \times 0.26 = 2.444$

40) $5.08 \times 7 = 35.56$

41) $2.35 \times 1.6 = 3.76$

42) $8.6 \times 4.4 = 37.84$

43) $0.94 \times 5.5 = 5.17$

44) $14.2 \times 0.7 = 9.94$

45) $3.75 \times 2.4 = 9$

46) $6.05 \times 1.2 = 7.26$

Decimal Operations: Practice (continued)

Part 5 Make the divisor whole Multiply both by 10 or 100. Then divide.

47) $4.68 \div 0.6 =$ **46.8** $\div 6 =$ **7.8**

48) $7.35 \div 0.5 =$ **73.5** $\div 5 =$ **14.7**

49) $2.16 \div 0.08 =$ **216** $\div 8 =$ **27**

50) $9.45 \div 0.7 =$ **94.5** $\div 7 =$ **13.5**

51) $1.56 \div 0.12 =$ **156** $\div 12 =$ **13**

52) $11.7 \div 0.9 =$ **117** $\div 9 =$ **13**

53) $3.24 \div 0.04 =$ **324** $\div 4 =$ **81**

54) $6.75 \div 1.5 =$ **67.5** $\div 15 =$ **4.5**

55) $0.928 \div 0.4 =$ **9.28** $\div 4 =$ **2.32**

56) $18.2 \div 2.6 =$ **182** $\div 26 =$ **7**

Decimal Operations: Practice (continued)

Part 6 Divide No help this time.

57) $9.36 \div 4 = 2.34$

58) $27.6 \div 0.6 = 46$

59) $8.05 \div 0.7 = 11.5$

60) $15.84 \div 3.6 = 4.4$

61) $4.5 \div 0.09 = 50$

62) $31.2 \div 8 = 3.9$

63) $6.39 \div 0.3 = 21.3$

64) $2.94 \div 0.42 = 7$

65) $50.4 \div 1.2 = 42$

66) $7.56 \div 6 = 1.26$

67) $1.44 \div 0.18 = 8$

68) $22.5 \div 0.05 = 450$

Decimal Operations: Practice (continued)

Part 7 More multiplying

69) $5.5 \times 7.47 = 41.085$

70) $3.5 \times 3.72 = 13.02$

71) $2.4 \times 8.38 = 20.112$

72) $4.4 \times 8.52 = 37.488$

73) $1.6 \times 4.44 = 7.104$

74) $3.3 \times 1.74 = 5.742$

75) $5.1 \times 3.72 = 18.972$

76) $7.7 \times 9.08 = 69.916$

77) $3.5 \times 8.63 = 30.205$

78) $2.2 \times 1.06 = 2.332$

79) $3.9 \times 4.66 = 18.174$

80) $6.9 \times 5.43 = 37.467$

Decimal Operations: Practice (continued)

Part 7 More multiplying (continued)

81) $2.1 \times 9.47 =$ **19.887**

82) $5.4 \times 7.94 =$ **42.876**

83) $9.7 \times 1.08 =$ **10.476**

84) $5.7 \times 9.27 =$ **52.839**

85) $1.2 \times 9.38 =$ **11.256**

86) $5.1 \times 9.23 =$ **47.073**

Part 8 More dividing

87) $3.55 \div 0.5 =$ **7.1**

88) $1.36 \div 0.8 =$ **1.7**

89) $4.56 \div 0.8 =$ **5.7**

90) $1.32 \div 0.2 =$ **6.6**

91) $4.08 \div 0.6 =$ **6.8**

92) $7.11 \div 0.9 =$ **7.9**

Decimal Operations: Practice (continued)

Part 8 More dividing (continued)

93) $1.74 \div 0.2 = 8.7$

94) $2.92 \div 0.4 = 7.3$

95) $1.02 \div 0.2 = 5.1$

96) $2.72 \div 0.8 = 3.4$

97) $0.48 \div 0.2 = 2.4$

98) $1.3 \div 0.2 = 6.5$

99) $0.45 \div 0.3 = 1.5$

100) $4.27 \div 0.7 = 6.1$

101) $0.6 \div 0.3 = 2$

102) $2.52 \div 0.3 = 8.4$

103) $5.64 \div 0.6 = 9.4$

104) $1.24 \div 0.2 = 6.2$

Decimal Operations: Practice (continued)

Part 9 Remember? Problems from earlier packets.

105) $4{,}872 \div 24 =$ **203**

106) 40% of 85 = **34**

107) $\frac{2}{3} \div \frac{1}{9} =$ **6**

108) 7 : 3 = **28** : 12

109) 90 km in 6 hours = **15** km per hour

110) $(7 + 8) \times 6 =$ **90**

Part 10 Word problems Line up the points. Name the pieces.

111) Mia buys 3.5 pounds of apples at $2.40 a pound. How much does she pay?

Number sentence **3.5 × $2.40 = $8.40**

Answer **$8.40**

Draw or show your work

112) A ribbon is 7.2 m long. It is cut into pieces 0.6 m long. How many pieces?

Number sentence **7.2 ÷ 0.6 = 12**

Answer **12** pieces

Draw or show your work

113) Leo ran 4.75 km on Monday and 6.8 km on Tuesday. How far did he run in all?

Number sentence **4.75 + 6.8 = 11.55**

Answer **11.55** km

Draw or show your work

Decimal Operations: Practice (continued)

Part 10 Word problems (continued)

114) A jug holds 3.2 liters. Ava pours out 1.45 liters. How much is left?

Number sentence **3.2 − 1.45 = 1.75**

Answer **1.75** liters

Draw or show your work

115) Challenge: Pens cost $1.35 each. How many can you buy with $12.15?

Number sentence **$12.15 ÷ $1.35 = 9**

Answer **9** pens

Draw or show your work

Check-In: Decimal Operations

Show what you know. No help this time.

Mastery: 6 of 7 (80%)

Part 1 Add and subtract

1) $82.3 - 19.56 =$ **62.74**

2) $26.07 - 8.9 =$ **17.17**

Part 2 Multiply

3) $0.38 \times 2.5 =$ **0.95**

4) $1.8 \times 6.3 =$ **11.34**

Part 3 Divide

5) $2.88 \div 0.24 =$ **12**

6) $8.64 \div 0.8 =$ **10.8**

Part 4 Word problem

7) Grapes cost $3.20 a pound. Sam buys 2.5 pounds. How much does he pay?

Answer **$8**

2.5 × $3.20 = $8